

Literature Review

AI-Driven Automated Interview Platform with Avatar-Based Candidate Assessment

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The proposed system is an AI-driven automated interview platform that combines a personalized AI avatar, LLM-based question generation and response evaluation, speech processing, candidate ranking, privacy-aware phone-use detection, and a deterministic fallback mechanism.

Scopus / SCI-Oriented Literature Review

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Suen, Chen & Lu (2019)	AI and Asynchronous Video Interviews (AVI)	Examined synchronous versus asynchronous video interviews and AI decision agents, including effects on interview ratings and applicant attitudes. The study reports that AVI can reduce some influence of first impressions and physical appearance.	Candidates showed less favourable reactions to AVI in some conditions. The work does not provide an end-to-end LLM interview platform or personalized interviewer avatar.	St te a ri A
Köchling et al. (2021)	AI-based video interviewing and recruitment	Research on AI video interviews identifies potential benefits including time savings, cost reduction, consistency and scalability, while also discussing applicant concerns about bias and accountability.	Concerns remain regarding algorithmic accountability, digital inequality and candidate perceptions. The study does not implement a complete technical solution for these issues.	St or m pa sig
Suen & Hung (2023)	Trust in AI-based video interviews	Investigated how tangibility, immediacy and transparency of AI interfaces affect applicants' trust in asynchronous video interviews.	Focuses primarily on trust and interface characteristics rather than comprehensive LLM-based question generation, assessment and ranking.	Pr fo in as
Suen & Hung (2024)	AI interfaces and applicant behaviour	Studied how different AI interface characteristics affect honest/deceptive impression management during asynchronous video interviews.	Shows that interface design can influence candidate behaviour; it does not develop a complete automated assessment pipeline.	R of in ac tic
Pentland, Wang & Twyman (2024)	Automated assessment using asynchronous video interviews	Developed a structured approach for extracting behavioural signals from AVI recordings for job-performance assessment and emphasized valid behavioural cues and unbiased training labels.	Stronger benchmarks and validation are needed before automated AVI assessment can be relied upon for employment decisions.	D va di ag in

Fisher, Bonaccio & Connelly (2024)	AI selection tools and applicants with disabilities	Examined opportunities and barriers created by AI-based selection tools, including asynchronous video interviews, with emphasis on reliability, validity and fairness.	Disability and accommodation considerations remain insufficiently integrated into AI selection design and validation.	Stalder et al. (2024) discuss the importance of considering individual differences and the need for personalized AI-based selection tools.
González-Morales, Martín-López & Góngora-Larequi (2026)	Systematic review of AI in recruitment and personnel selection	Scopus-based review covering publications from 2020 to February 2025. Identifies AI applications in candidate screening, interview assessment, soft-skill assessment, CV analysis and recruitment automation.	The field remains relatively nascent and dispersed, with continuing need for validation, fairness research and practical deployment evidence.	Pirsa et al. (2026) discuss the importance of considering individual differences and the need for personalized AI-based selection tools.
Dadaboyev et al. (2025)	Systematic review of AI in employee recruitment	Reviewed peer-reviewed literature and identified automated CV parsing, chatbot screening, predictive job-fit analysis and AI-assisted video interview analysis as major applications.	Highlights continuing challenges involving algorithmic bias, transparency, legal compliance and industry-specific implementation.	Stalder et al. (2024) discuss the importance of considering individual differences and the need for personalized AI-based selection tools.
Piras et al. (2026)	Multi-agent LLM-based recruitment and human-AI interaction	Introduced a multi-agent LLM recruitment platform focused on scalability, fairness, transparency and robustness, including semi-structured interviews and multilingual/multichannel input.	Evaluation involved a relatively small participant group, leaving room for larger controlled studies and broader validation.	Pirsa et al. (2026) discuss the importance of considering individual differences and the need for personalized AI-based selection tools.

Key Research Gaps

1. Integration Gap

Existing studies frequently investigate individual components such as asynchronous video interviewing, applicant trust, AI scoring, or fairness. There is comparatively less work integrating AI avatar interaction, STT/TTS, LLM question generation, multi-dimensional scoring, ranking and resilience into one system.

2. Validation Gap

Automated interview scores require comparison with expert human interviewer ratings. The proposed system currently has functional and qualitative evaluation but has not yet established concurrent or predictive validity.

3. Fairness Gap

Demographic fairness and scoring parity need systematic evaluation before automated assessment is used in production hiring.

4. Accessibility Gap

AI interviews should support captioning, alternative response modes and auditable accommodations for candidates with disabilities.

5. Transparency and Trust Gap

Research indicates that transparency and interface characteristics can influence applicant trust and behaviour. The assessment process should therefore make evaluation dimensions understandable.

6. Reliability Gap

External LLM APIs introduce outage and rate-limit risks. The proposed deterministic fallback provides a practical resilience mechanism, although it sacrifices some semantic richness.

7. Multimodal Assessment Gap

Future research can investigate calibrated fusion of transcript, audio and video signals while explicitly testing fairness impacts.

Relevance to the Proposed System

The proposed platform builds on the existing literature by combining a personalized AI interviewer avatar with role- and resume-tailored question generation, speech processing, five-dimensional response scoring (relevance, accuracy, confidence, keyword coverage and skill signal), candidate ranking, privacy-preserving phone-use flagging, and graceful degradation through deterministic fallbacks.

The proposed system treats candidate ranking as a decision-support signal for a human recruiter rather than an autonomous accept/reject mechanism. This provides a basis for integrating AI automation while retaining human oversight.

Literature Review Conclusion

Existing literature demonstrates that AI and asynchronous video interviewing can improve recruitment scalability and consistency, but important challenges remain in assessment validity, fairness, transparency, accessibility and candidate trust.

The proposed system addresses a systems-integration gap by bringing several capabilities together in one platform. Its principal research opportunity is to move from a functional systems contribution toward empirically validated assessment through expert-rating comparison, inter-rater reliability analysis, demographic fairness auditing and multimodal evaluation.

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